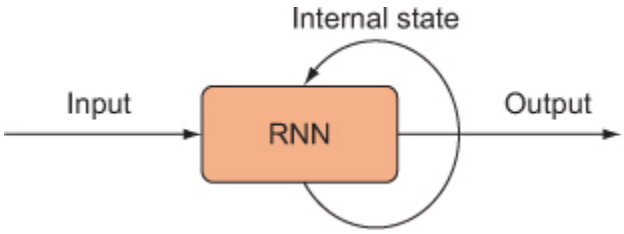
English	Chinese
I ate food.	我吃饭了.

Each word, w_i, in English is encoded as a fixed-length one-hot vector, $w_i: \mathbb{B}^n$, with dimensionality n. The dimensionality determines the maximal vocabulary size. For example, if $n = 10$, the model can only handle a vocabulary of 10 words in total, so usually it is much larger, such as $n \approx 40000$. Likewise, each word in Chinese is encoded as a fixed-length vector. We want to build a translation model that can translate each word of English into Chinese.

The first approaches to this problem were based on recurrent neural networks, which are inherently sequential models, as they are capable of storing data from each input. A recurrent neural network, at a high level, is a function that maintains an internal state that is updated with each input that it sees (figure 10.6).

Figure 10.6. A recurrent neural network (RNN) is capable of maintaining an internal state that is updated with each new input it receives. This allows RNNs to model sequential data such as time series or language.



Most RNN language models work by first having an encoder model that consumes a single English word at a time, and once done gives its internal state vector to a different decoder RNN that outputs individual Chinese words one at a time. The problem with RNNs is that they are not easily parallelized because you must maintain an internal state, which depends on the sequence length (figure 10.7). If sequence lengths vary across inputs and outputs, you have to synchronize all the sequences until they’re done processing.